

eQRAP Expedice (Lite): admin guide

This is the simple version of the app, built for **warehouse and expedice**, not production. It's two files: `Code.gs` and `Index.html`. You manage almost everything in the Google Sheet, so you don't need to open the code.

The form follows the firm's QRAP V3.0 form. Sections ① to ⑦ ask the same questions and follow the same QRQC rules as the original eQRAP (logistics version). Only the parts that belong to production or APU were removed: the APU and plant-level copies of a QRAP.

1. What changed compared with the big version

	Big version (apps-script/)	Lite (eqrap-lite/)
Files to paste	39	2 (+ optional <code>appsscript.json</code>)
Lines of code	~7,200	~2,750 (≈1,300 server, ≈1,450 screen incl. styles)
Questions ①–⑦	QRAP V3.0 (logistics)	the same , without the APU / plant level
Roles to set up	10 (kiosk, operator, TL, supervisor, pilot, quality, EHS, manager, APU manager, admin)	2 in the tab Lidé: VEDOUCÍ and MANAŽER. Everyone else is a normal user; pilots are chosen in ④
Screens	13	4 : Tabule · Nový QRAP · Detail (form ①–⑦) · Jak to funguje
Sheet tabs	~20	8 (you edit only 3 of them)
Waiting	each screen asked the server (1–3 s)	only first opening and saving wait. Moving around is instant

Why it's faster: when the page opens, it receives all open problems at once. Clicking a problem, filtering and searching then happen in the browser with no server call. The server is only called when you **save**, plus a quiet refresh every 90 seconds.

What stays exactly as on the form:

Section	Questions and rules
① POPIS PROBLÉMU	SAFETY · CO je za problém? · JAK byl objeven? (+ jiné) · KDY? · KDE? zóna + lokace · Kde vzniklo? · KOLIK? + jednotka / NOK situace bez kusů · Stejný problém v posledních 7 dnech? (+ číslo QRAP) · JMÉNO nebo číslo odznaku · číslo materiálu, HU / dodacího listu, dodavatel · Foto ŠPATNĚ + Foto SPRÁVNĚ (or an exception with a reason)
② Kdo byl upozorněn?	roles (+ jiné). Kvalita is added automatically when ③ risk = ANO
③ OKAMŽITÁ OPATŘENÍ <24h	Je riziko u zboží na skladě / expedovaného? (ANO → zkontrolováno / nalezeno špatně / kontrola probíhá, Kvalita gets an e-mail) · list of immediate actions (Typ, Co, Kdo, Hotovo v) · Proces zastaven? (+ obnoven v). ③ is complete only when every rule is met; the TL may finish it later
④ Rozhodnutí OJT	Informace všem směnám (5' meeting) · Pokračovat s analýzou a 5 Proč (+ piloti) / Problém vyřešen: ŽÁDNÉ 5 Proč (not allowed when SAFETY = ANO) / Eskalace · komentář · signature. After signing, ①–④ are locked

Section	Questions and rules
⑤ PŘÍČINA 5 Proč?	Proč to vzniklo? (5×) · Proč to nebylo zachyceno? (5×) · kořenová příčina · Co jsme se naučili z QR? · Příležitost k aktualizaci STANDARDU? (ANO creates the action "Aktualizovat standard")
⑥ Konečné akce	each action removes one 5 Proč step, has a pilot, plan, done date and evidence link · efektivita na 5 různých směnách ("neefektivní" sends the QRAP back to analysis and a new action is needed) · closure checklist + signature
⑦ Hodnocení QRQC	1 RED BOX · 2 ROZDĚLENÍ · 3 POROVNÁNÍ · 4 POUČENÍ A SDÍLENÍ · 5 ON JOB TRAINING (OK / NOK / N/A) · OJT s kým · zpětná vazba + typ. A QRAP can be assessed many times

What was left out:

- **Removed because it's APU / production only:** APU and plant-level QRAP copies.
- **Simplified app features (the form questions are not affected):**
 - drafts and the step-by-step wizard (now one form)
 - the manager dashboard
 - English
 - notes with review dates on actions
 - per-shift e-mail routing
 - 11 e-mail types reduced to 7 plus an optional morning reminder

2. Setup (about 10 minutes, one time)

If you already set up an earlier Lite test, start again with a **new Sheet**. The columns changed, so the old test tabs don't fit.

1. Create a **new Google Sheet**, for example "eQRAP Expedice". Use the account that should own the app (see §7 about ownership).
2. In the Sheet, open **Extensions** → **Apps Script**.
 - In `Code.gs`, delete everything and paste the content of `eqrap-lite/Code.gs`.
 - Click **+** → **HTML** and name it `Index`. Delete everything and paste `eqrap-lite/Index.html`.
 - (Optional) In ⚙ Project Settings, tick "Show appsscript.json" and paste `eqrap-lite/appsscript.json`. If you skip this, set the time zone to *Europe/Prague* in Project Settings.
 - Click  Save.
3. In the toolbar, choose the function `setup`, click **Run** and allow the permissions. This creates the 8 tabs and a Drive folder "eQRAP fotky".
4. Go back to the Sheet and reload it. A new menu **eQRAP** appears. Then:
 - In **Lidé**, replace the example people with real people (see §4).
 - In **Seznamy**, check the zones, shifts and lists.
 - In **Nastavení**, fill in `EMAIL_KVALITA` and `EMAIL_BOZP`, and set `EMAILY` to `NE` while you are testing.
5. Back in Apps Script, click **Deploy** → **New deployment** → ⚙ **Web app**:
 - *Execute as:* **Me**
 - *Who has access:* **Anyone within [your company]**
 - Click **Deploy** and copy the URL. That is the app.
 - For the kiosk PC, add `?kiosk=1` to the end of the URL.

6. (Optional) Use **eQRAP** → **Vložit ukázková data** to see 3 example QRAPs. It's better to try this in a separate test Sheet.

Updating the code later: paste the new code, run **eQRAP** → **Nastavit tabulku** once (it adds any new columns), then go to **Deploy** → **Manage deployments** →  → **Version: New version** → **Deploy**. The URL stays the same.

3. How it works: the flow and who does what

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①②③ QR – operátor / TL (③ do 24 h od KDY)
↓
④ Rozhodnutí OJT – vedoucí podepíše
├ „Problém vyřešen: ŽÁDNÉ 5 Proč“ → uzavřeno hned
├ „Eskalace“ → manažer dostane e-mail, analýza pokračuje
↓
⑤ 5 Proč – piloti
⑥ Konečné akce – piloti akcí
⑥ Efektivita na 5 různých směnách – TL / kdokoli
⑥ Uzavření – vedoucí podepíše
⑦ Hodnocení QRQC – vedoucí / manažer (kdykoli, i vícekrát)

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The **status** is **calculated automatically** from what has been filled in. Nobody sets it by hand:

Status in the app	Means	Who acts
③ Opatření	sent, but ③ is not complete yet (the page lists what is missing; deadline 24 h from KDY)	anyone on the shift (TL)
④ Rozhodnutí	③ complete, waiting for the signed decision	VEDOUcí
⑤ 5 Proč	root cause not marked yet, or no action yet (after "neefektivní": no new action yet)	pilots
⑥ Akce	definitive actions are open	pilots of the actions
⑥ Ověření 5 směn	all actions done: record 5 effective shifts, then the leader signs the closure	TL / anyone + VEDOUcí
Uzavřeno / Zrušeno	finished / cancelled by a manager (the record stays)	–

Roles. You set up only two roles in Lidé. Pilots are chosen for each QRAP in ④.

	Everyone with a company account (no row in Lidé needed)	Pilot (chosen in ④)	VEDOUcí	MANAŽER
①②③ new QRAP (③: only the risk answer is needed at first)	✓	✓	✓	✓
③ finish the immediate actions (until ④ is signed)	✓	✓	✓	✓
①② edit the description (until ④ is signed)	only their own QRAP	only their own QRAP	✓	✓
④ tick informed shifts, sign the decision	–	–	✓	✓

	Everyone with a company account (no row in Lidé needed)	Pilot (chosen in ④)	VEDOUCÍ	MANAŽER
⑤ 5 Proč · ⑥ definitive actions	–	✓	✓	✓
⑥ mark own action done	✓	✓	✓	✓
⑥ record a 5-shift slot	✓	✓	✓	✓
⑥ sign the closure · ⑦ QRQC assessment	–	–	✓	✓
cancel / reopen / unlock ①–④	–	–	–	✓

The person who owns the script is always MANAŽER.

Who gets e-mails:

When	Who
new QRAP	leaders of that zone (empty zone in Lidé = all zones)
SAFETY = ANO	EMAIL_BOZP + managers
③ risk = ANO	EMAIL_KVALITA
④ pilots chosen	the pilots
④ Eskalace	managers
new definitive action	the pilot of the action
⑥ "neefektivní"	the pilots + leaders of the zone
every morning (optional)	reminder to leaders about open QRAPs, and to pilots about late actions

What each person sees. Everyone sees the **same 4 screens**, and buttons appear only if that person is allowed to use them. The **"Moje úkoly"** tile on the board shows each person only what is waiting for them. The **kiosk** (?kiosk=1) is the same app with 3 differences:

- JMÉNO is left empty, to be typed in
- a "QRAP odeslán" screen appears after sending
- it returns to the board by itself after 2 minutes without use

4. Everyday admin tasks, all in the Sheet (no code)

I want to...	Do this
add a person or make someone a leader	Lidé: add a row with Jméno, E-mail (exactly their Google account), Role VEDOUCÍ or MANAŽER , Oblast (= zone, empty = all)
let someone be a pilot	add them to Lidé (no role needed). The leader then picks them in ④
take a role away / someone left	Lidé: set Aktivní to NE (keep the row so the history still shows the name)
change zones, shifts or the lists in the form	Seznamy columns: Zóny · JAK byl objeven · Jednotky · Kdo byl upozorněn · Typy okamžitých opatření · Směny. Old QRAPs keep their old values

I want to...	Do this
set who gets Quality / safety e-mails	Nastavení → EMAIL_KVALITA , EMAIL_BOZP
change the 24 h deadline for ③	Nastavení → LHUTA_OPATRENI_HODIN
turn e-mails off or on	Nastavení → EMAILY = NE / ANO
send a morning reminder	menu eQRAP → Zapnout denní připomínky
rename the app	Nastavení → NAZEV
show closed QRAPs for longer on the board	Nastavení → ZOBRAZIT_UZAVRENE_DNI
correct ①–④ after the decision was signed	in the app, as manager: Správa → Odemknout ①–④ (with a reason). The decision must then be signed again
get rid of a duplicate or mistaken QRAP	in the app, as manager: Správa → Zrušit QRAP (with a reason). It stays in the history
reopen a closed QRAP	in the app, as manager: Správa → Znovu otevřít
make a report or chart	add your own tab with formulas (for example =COUNTIF(Problémy!B:B;"UZAVŘENO")), or File → Download → Excel
back up	File → Version history works automatically; also make a monthly File → Make a copy

Changes in Lidé, Seznamy and Nastavení apply **within 5 minutes**. To apply them right away, use **eQRAP** → **Použít změny v nastavení hned**. Users then reload the app.

Tabs in the Sheet

Tab	What it is	May I edit it?
Lidé	people, roles, zones	✓ yes, this is your main control
Seznamy	the dropdown values of the form	✓ yes (don't rename the column headers)
Nastavení	name, e-mails, deadlines	✓ only the <i>Hodnota</i> column
Problémy	one row = one QRAP (columns ① ... ⑥)	⚠ read only. Change data through the app
Akce	one row = one action (Část ③ = okamžité, ⑥ = konečné)	⚠ read only
Ověření	the 5-shift effectiveness slots (all rounds)	⚠ read only
Hodnocení	⑦ QRQC assessments	⚠ read only
Historie	who changed what and when	⚠ read only (this is your audit trail)

The five data tabs are protected with a warning, so you get a prompt before editing them by hand.

Never do these:

- rename, delete or reorder the columns in the data tabs (you may add your own columns *to the right*)
- delete the tabs Lidé, Seznamy or Nastavení
- give everyone edit access to the Sheet. Users don't need it at all, because the app works without it

5. If something is wrong

Problem	Fix
"Aplikaci se nepodařilo spustit – chybí list / sloupec"	run eQRAP → Nastavit tabulku (setup) . It repairs things and never deletes data. Check that no column header was renamed
a leader has no "Rozhodnout" button	③ must be complete first (the page shows what is missing). Their e-mail in Lidé must match their Google account exactly, and Aktivní must be ANO . Then eQRAP → Použít změny... and reload
a pilot can't edit ⑤	the pilot must be chosen in ④ (or added with "+ přidat pilota" by a leader)
"Podepsat uzavření" is greyed out	the checklist under it shows what is missing (e.g. "Co jsme se naučili z QR?" or 5 different effective shifts)
e-mails don't arrive	EMAILY = ANO , the person has an e-mail in Lidé / Nastavení, check spam. Google allows about 1,500 e-mails a day
someone gets "you need permission"	the deployment must be "Anyone within [company]" and they must be signed in with their company Google account
I changed the code but nothing changed	Manage deployments → New version (\$2)
the first opening takes 2–4 s	normal: Google starts the script. After that, clicks are instant and saves take about 1–2 s

6. If you ever need to look at the code

Both files are split into numbered sections with a table of contents at the top. Typical changes:

Change	Where
texts of the form (labels, hints)	<code>Index.html</code> section 7 (<code>popisForm</code> , <code>opatreniForm</code> , <code>decisionForm</code> , <code>pricinaForm</code> , <code>actionForm</code> , <code>assessForm</code>)
what a form part checks	<code>Code.gs</code> section 6 (<code>popisFrom_</code> , <code>opatreniFrom_</code> , <code>pricinaFrom_</code> , <code>defActionFields_</code>)
who may do what	<code>Code.gs</code> section 5: one-line rules <code>canPopis_</code> ... <code>canAssess_</code>
when a status changes, ③ completeness, closure checklist	<code>Code.gs</code> section 4 (<code>statusOf_</code> , <code>s3Missing_</code> , <code>checklist_</code>)
e-mail texts	<code>Code.gs</code> section 7 (<code>mailNew_</code> , <code>mailQuality_</code> , <code>mailPilots_</code> ...)
colours	<code>Index.html</code> top of <code><style></code> (<code>--accent</code> etc.)
a new column	<code>Code.gs</code> section 1 <code>COLS</code> , then run setup (it adds the missing header)

Code.gs (server)	Index.html (screen)
1 tabs, columns, defaults	1 state and texts
2 web app entry (<code>doGet</code>)	2 small helpers (<code>h</code> , <code>api</code> , <code>save</code>)
3 API called by the page	3 header and routing
4 status rules	4 board (Tabule)

Code.gs (server)	Index.html (screen)
5 permissions	5 new QRAP
6 sheet helpers and form checks	6 QRAP detail ①–⑦
7 e-mails, daily reminder	7 forms ① ... ⑦
8 setup, menu, example data	8 photos · 9 help, kiosk, refresh

7. Is it safe to use in the firm?

Short answer: it's safe for an **internal pilot in expedice**. Before it becomes the official QRAP record, get IT and Quality to approve it and move the ownership to a team account.

What protects it

- **Only company accounts** can open it ("Anyone within [company]"). People outside the company can't.
- **Users never get access to the Sheet or the Drive folder.** The app runs as the owner, and **every save is checked on the server** against the Lidé tab and the QRAP rules. Hiding a button in the browser is only for convenience. The real check is on the server, so a user can't bypass it.
- **Admin functions** (setup, example data, reminders) run only for the owner of the script. Nobody can start them from the browser.
- **Everything is logged** in Historie: who, when and what. Nothing can be deleted from the app. "Zrušit" keeps the record with a reason, and removed actions stay in the Sheet marked "Odebráno".
- **Signed parts are locked.** After ④ is signed, ①–④ can only be unlocked by a manager, with a reason that goes to the history.
- **Typed text can't do harm.** It's always shown as plain text, never run as code. In the Sheet it's stored as text, never as a formula, so values like "00123" stay exactly as typed. Evidence links must start with http(s).
- A photo can be opened **only through its own QRAP**, so nobody can read other files from the owner's Drive.
- **No outside services or libraries.** Everything stays in your company's Google Workspace, the same place as your e-mail and Drive.
- **Tested:** 176 automatic server checks (every rule of ①–⑦, permissions, e-mails, bad input) and a full browser test of the whole flow from reporting to closure and assessment, including the kiosk and a phone.

Risks you should know about (and what to do)

Risk	What to do
It depends on one person's account. If the owner leaves and the account is deleted, the app stops and the photos disappear	Put the Sheet in a Shared Drive , or create it under a team account (e.g. qrqc-expedice@...). Have 2 people with edit access to the Sheet
It isn't a validated system (IATF 16949 / customer requirements)	Run it in parallel with the paper QRAP for 2–4 weeks . Let Quality decide if it can replace the paper
Everyone in the company can see all QRAPs	This is intended for transparency. Don't write health details of injuries into it; BOZP injuries still go to the official injury book
Kiosk shared account: JMÉNO is typed, not verified	Never give the kiosk account a role. Signatures (④, closure) are made from personal accounts

Risk	What to do
Google limits: about 1,500 e-mails a day; about 30 people saving <i>at the same second</i>	Plenty for one warehouse. Not meant for the whole company
Personal data (names, e-mails of employees)	Tell IT / the data protection officer. It stays inside your Google Workspace
Backups	Version history is automatic. Also make a monthly File → Make a copy

Pilot checklist

- ☐ IT agrees with an Apps Script web app on the company domain (and with a kiosk account, if used).
- ☐ Owner is a team account or Shared Drive; 2 admins have edit access to the Sheet.
- ☐ Lidé filled in: every leader and manager, and everyone who can be a pilot. E-mails checked.
- ☐ Seznamy match your zones, shifts and lists; EMAIL_KVALITA and EMAIL_BOZP filled in.
- ☐ One week with EMAILY = NE on a test Sheet, then switch to the real Sheet with EMAILY = ANO .
- ☐ Short training (5 min): show "Jak to funguje" in the app.
- ☐ 2–4 weeks in parallel with paper, then Quality decides.

Placeholder data only: all example people use example.com addresses.